BUSINESS ANALYTICS SEMINAR

Employee Attrition Prediction

Turning historical HR data into early warning signals — and targeted retention action

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1 HR Data

2 Preprocess

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4 EDA

5 ML Models

6 Prediction

7 Insights

8 Retention

HR Data · Preprocessing · Feature Engineering · EDA · Machine Learning · Prediction · Business Insights · Retention Actions

MOTIVATION

Why Employee Attrition Analysis Matters

Losing the wrong employee at the wrong time costs far more than a vacant seat

Hiring & Training Investment

Organizations commit significant time and money to recruit and train every employee.

Knowledge & Productivity Loss

When experienced employees leave, their institutional knowledge and output leave with them.

Understanding Workforce Patterns

Attrition analysis helps HR teams see the trends behind who leaves, and why.

Focused Retention Effort

Insight lets HR concentrate energy on the employees and teams that need it most.

STAGE 1 · FOUNDATION

The Dataset

HR Data | Preprocess | Feature Eng. | EDA | ML Models | Prediction | Insights | Retention

14,999

Employee Records

10

Variables per Record

- Department
- Role
- Tenure
- Salary Level
- Satisfaction
- Performance
- Monthly Hours
- Promotions

STAGE 2 · CLEANING

Data Preprocessing

HR Data · Preprocess · Feature Eng. · EDA · ML Models · Prediction · Insights · Retention

01

Check for Missing Values

Scan every field for gaps that could bias the model.

02

Flag Inconsistent Records

Identify entries that conflict with expected ranges or logic.

03

Remove Duplicates

Drop repeated employee entries that would double-count patterns.

04

Encode Categorical Variables

Convert department, role, and salary level into model-readable form.

05

Prepare Numerical Variables

Scale and structure tenure, hours, and satisfaction scores.

06

Create Useful Features

Derive analysis-ready fields from the cleaned raw data.

STAGE 3 · SIGNAL DESIGN

Feature Engineering

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Tenure

Years the employee has worked at the company.

Satisfaction Level

Employee satisfaction score from survey data.

Performance Rating

Latest performance evaluation result.

Avg. Monthly Hours

Average working hours per month.

Salary Level

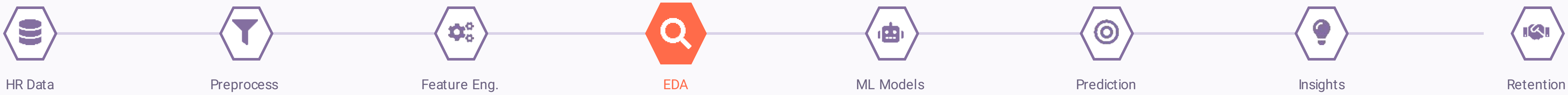
Compensation category — low, medium, or high.

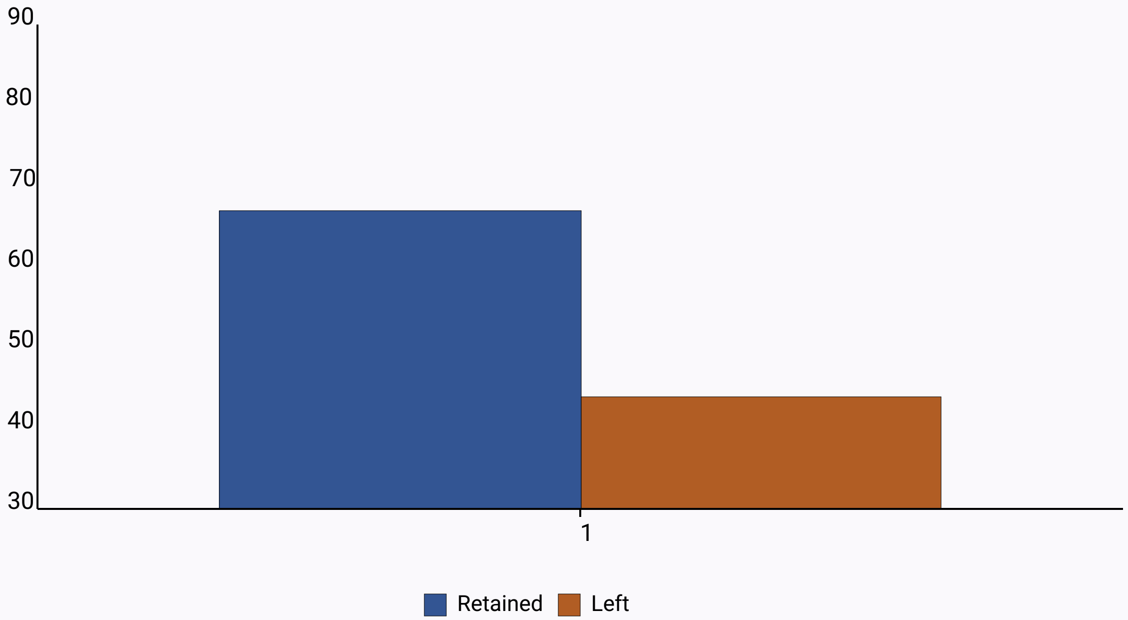
Department / Role

The employee's job function and team.

STAGE 4 · EXPLORATION

Exploratory Data Analysis (EDA)





Illustrative pattern — directional, based on project findings

What the Analysis Covered

- Tenure trends across the workforce
- Satisfaction score distributions
- Performance rating patterns
- Workload (monthly hours) differences
- Retained vs. departed employee comparisons

Key finding: satisfaction has a strong relationship with attrition

STAGE 5 · MODELING

Machine Learning Models

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Baseline · Interpretable

Logistic Regression

An interpretable model used for binary classification. It predicts whether an employee belongs to the Retained (0) or Left (1) class.

Best Performer

Random Forest

Combines multiple decision trees and captures nonlinear relationships. In this project, it produced stronger predictive performance than Logistic Regression.

STAGE 6 · EVALUATION

Model Performance

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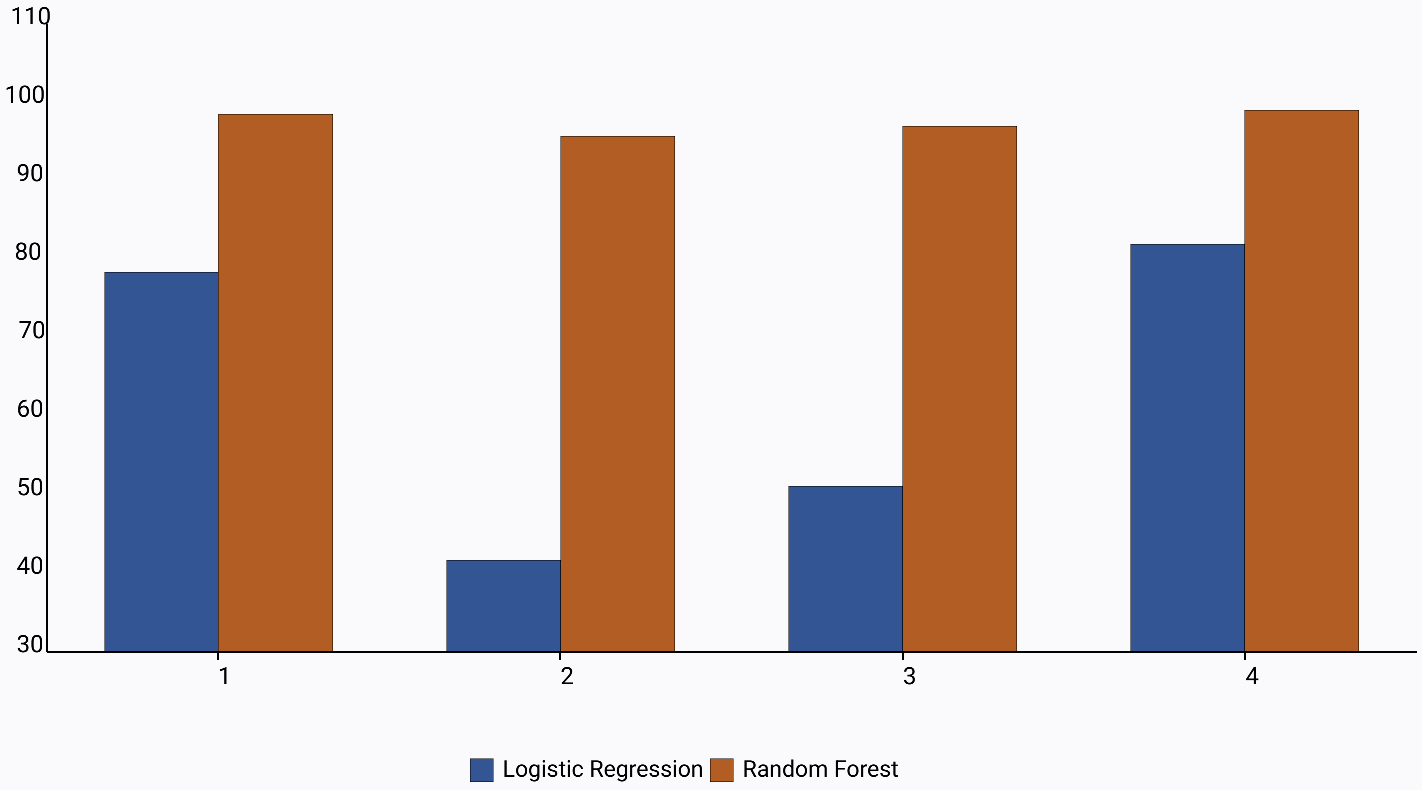


The Verdict

Random Forest performed substantially better on the current dataset across every metric.

98.57%

Random Forest Accuracy

99.10%

Random Forest ROC-AUC

STAGE 7 · SO WHAT?

Business Insights

HR Data · Preprocess · Feature Eng. · EDA · ML Models · Prediction · Insights · Retention

Satisfaction Is the Strongest Signal

Employee satisfaction shows the clearest relationship with attrition of any variable studied.

Tenure Links to Performance

Tenure and performance rating show a meaningful relationship worth monitoring over time.

Risk Varies by Group

Some departments and salary bands show noticeably higher attrition than others.

Early Identification Is Possible

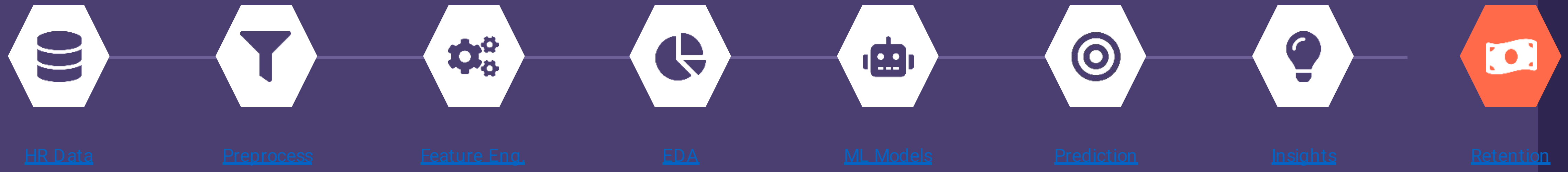
High-risk employees can potentially be flagged before they decide to leave.

STAGE 8 · CLOSING THE LOOP

Conclusion

Business Analytics turns raw HR records into decisions that keep good people from leaving.

The Full Journey



Preprocessing · EDA · Machine Learning · Prediction — combined into one retention strategy

Thank you